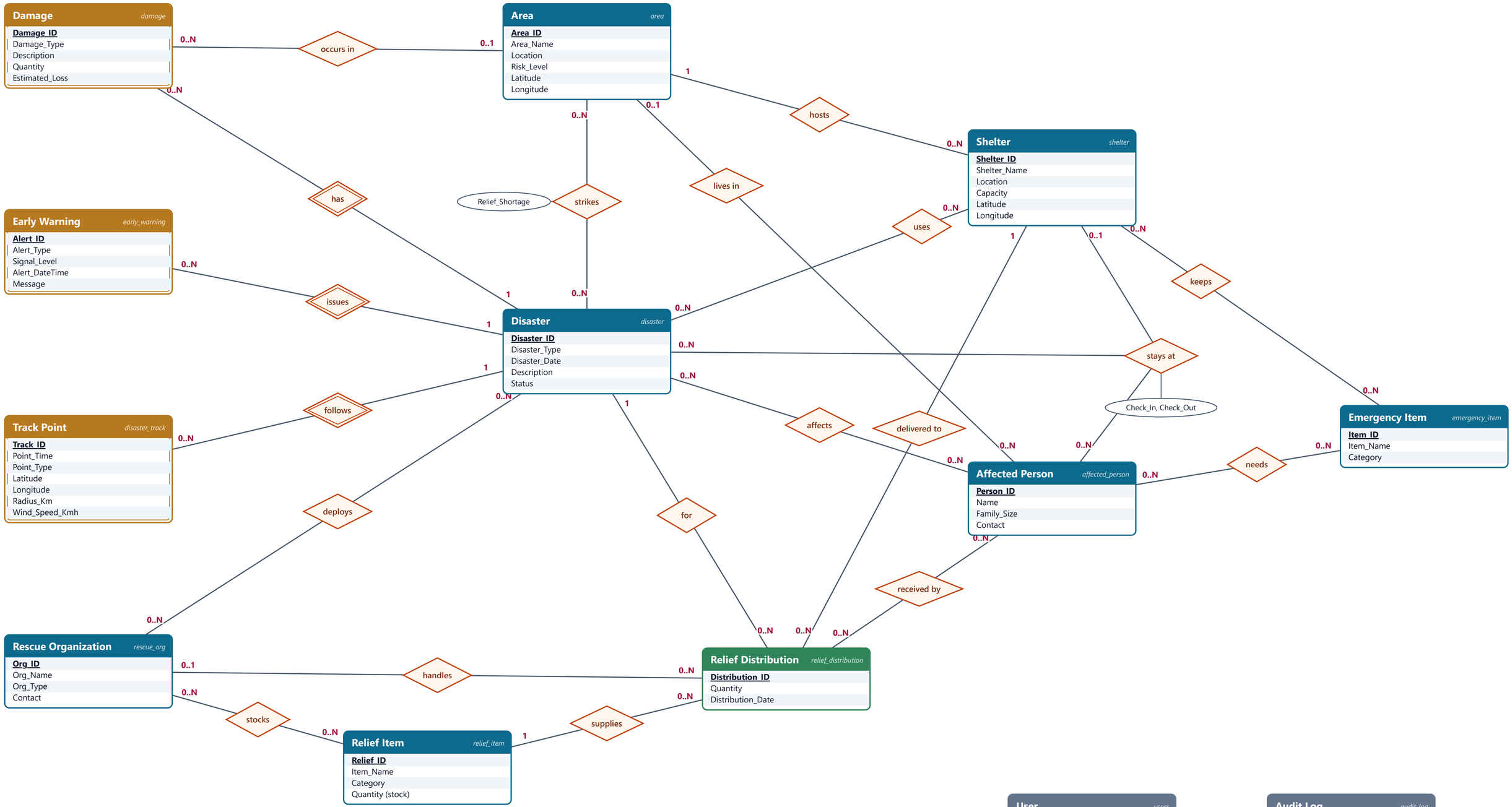
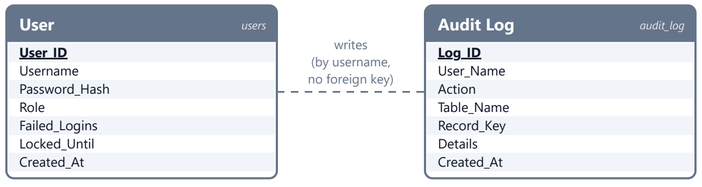
Disaster Mapping & Relief Management System — Conceptual ER Diagram

13 entities and 19 named relationships, derived from the 22-table MySQL schema: the 8 link tables become many-to-many relationships and shelter_assignment becomes the three-way relationship “stays at”. Foreign-key columns are omitted from the entities because the relationships carry them.





How to read this diagram

- Entity
- Dependent (weak) entity: cannot exist without its disaster
- Event entity: a relief handed out, with its own identity
- Stand-alone entity: no relationship to the others
- Relationship (named by its verb); double diamond = identifying relationship of a dependent entity

Underlined = primary key · ellipse = attribute of a relationship · italic grey name = MySQL table

Cardinality labels

The label next to an entity says how many of that entity can be linked to ONE instance of the entity at the other end.

0..N zero or many 1 exactly one 0..1 zero or one (the link is optional).

Example: “Area hosts Shelter”: a shelter belongs to exactly 1 area; an area hosts 0..N shelters.

“stays at” is a three-way relationship: for one person in one disaster there is at most 0..1 shelter.

Rules the relationships cannot show (enforced by database triggers)

- A stay needs a shelter used for that disaster and a person affected by that disaster.
- A distribution needs a shelter and (if given) an organization that took part in that disaster.
- A recipient must be a person affected by the disaster of that distribution.
- A damage record’s area, if given, must be one of the areas struck by that disaster.
- Shelter capacity and relief stock limits are checked by the application (inc/rules.php).

How each relationship is stored in MySQL

Every relationship in the conceptual diagram maps to exactly one foreign key (one-to-many) or one link table (many-to-many), 29 foreign keys in total.

#	Relationship	Between	Cardinality	Implemented as	ON DELETE
1	**strikes**	Disaster – Area	M : N · attribute Relief_Shortage	`link table disaster_area`	CASCADE / CASCADE
2	**has**	Disaster – Damage	1 : N · identifying	`damage.Disaster_ID`	CASCADE
3	**occurs in**	Area – Damage	1 : N · optional	`damage.Area_ID (NULL allowed)`	SET NULL
4	**issues**	Disaster – Early Warning	1 : N · identifying	`early_warning.Disaster_ID`	CASCADE
5	**follows**	Disaster – Track Point	1 : N · identifying	`disaster_track.Disaster_ID`	CASCADE
6	**uses**	Disaster – Shelter	M : N	`link table disaster_shelter`	CASCADE / CASCADE
7	**deploys**	Disaster – Rescue Organization	M : N	`link table disaster_rescue_org`	CASCADE / CASCADE
8	**affects**	Disaster – Affected Person	M : N	`link table disaster_affected_person`	CASCADE / CASCADE
9	**hosts**	Area – Shelter	1 : N	`shelter.Area_ID`	RESTRICT
10	**lives in**	Area – Affected Person	1 : N · optional	`affected_person.Area_ID (NULL allowed)`	SET NULL
11	**stays at**	Disaster – Person – Shelter	three-way · one shelter per person per disaster · attributes Check_In, Check_Out	`shelter_assignment, unique (Disaster_ID, Person_ID)`	CASCADE disaster · CASCADE person · RESTRICT shelter
12	**for**	Disaster – Relief Distribution	1 : N	`relief_distribution.Disaster_ID`	CASCADE
13	**delivered to**	Shelter – Relief Distribution	1 : N	`relief_distribution.Shelter_ID`	RESTRICT
14	**supplies**	Relief Item – Relief Distribution	1 : N	`relief_distribution.Relief_ID`	RESTRICT
15	**handles**	Rescue Organization – Relief Distribution	1 : N · optional	`relief_distribution.Org_ID (NULL allowed)`	SET NULL
16	**received by**	Relief Distribution – Affected Person	M : N	`link table relief_recipient`	CASCADE / CASCADE
17	**stocks**	Rescue Organization – Relief Item	M : N	`link table rescue_org_relief_item`	CASCADE / CASCADE
18	**needs**	Affected Person – Emergency Item	M : N	`link table affected_person_emergency_item`	CASCADE / CASCADE
19	**keeps**	Shelter – Emergency Item	M : N	`link table shelter_emergency_item`	CASCADE / CASCADE